NewsBot Intelligence System: Reflective Journal

ITAI 2373 Mid-Term Project | Kristina Akpan (individual submission) | October 17, 2026

1. Collaboration Analysis (Individual Project)

I was not assigned to a group, so I completed every module on my own. Without teammates to divide the work, I treated the eight modules as a single pipeline and planned the order of work around their dependencies: data preparation first, then preprocessing and TF-IDF, then the spaCy based analyses, and classification last, because the classifier reuses features from every earlier module. Working alone meant I had no one to review my code, so I built my own quality checks into the notebook: a fixed random seed, a held out test set that I touched only once, statistical tests for the POS differences, and a clean top to bottom run before submitting. As the assignment permits, I used an AI assistant to help me review and verifiy the outputs and numbers reported in the notebook.

2. Technical Integration Challenges

Choosing the right data. The Kaggle copy of the BBC dataset is entirely lowercase and has no headlines. That is fine for a bag of words classifier but breaks NER and POS tagging, which depend on capital letters. I switched to the original cased version of the same 2,225 article corpus and repaired an encoding error that had inserted a stray character before every pound sign.

Two kinds of text. The classifier needs cleaned, lemmatized text, while sentiment, parsing and NER need the original sentences with punctuation and capitals. I kept both versions side by side so each module used the right one.

Speed on the Colab free tier. Running spaCy separately for syntax, NER and features would have tripled the runtime, so I parsed all 2,000 articles once (about 90 seconds) and reused the parsed documents everywhere. A grid search over too many settings ran for more than ten minutes, so I trimmed it to the parameters that mattered. I also found a subtle bug: my parse tree depth function never ended, because spaCy creates a new token object each time you ask for a word's head, so comparing objects never matched. Comparing token positions fixed it.

Training and prediction must match. The starter template filled the extra features with zeros when classifying new articles, which silently gives the model different inputs than it was trained on. I wrote one feature function used for both training and prediction, and kept every step inside a scikit-learn Pipeline so the vocabulary was learned only from training data.

Results that did not match my expectations. VADER scores for whole articles were saturated near the extremes for 69% of articles, so I switched to the average sentence score. Adding my 26 engineered linguistic features slightly lowered accuracy, so I reported that honestly and kept the simpler TF-IDF model, which reached 96.75% test accuracy.

3. Business Value Assessment

NewsBot does the first pass of work that media teams currently do by hand. A newsroom could route 97% of articles to the right desk automatically and send only low confidence stories, such as the health article my system flagged at 32% confidence, to an editor. Entity level sentiment is directly useful for PR and reputation monitoring: the system identified WorldCom and Yukos, both at the center of scandals in 2004 and 2005, as the most negatively covered organizations without being told anything about them. For financial analysts, Business articles carried the most MONEY entities (2.8 per article), so dollar amounts and company names can feed a market monitoring dashboard. The main risks are that the training data comes from one outlet and one period, and that sentiment scores about real people and companies can be wrong, so a human should review them before any decision is made.

4. Individual Contributions

As the sole author I completed all components: the business case (Module 1); the preprocessing pipeline with contraction handling, unicode normalization and edge case tests (Module 2); TF-IDF parameter optimization, word clouds and chi squared analysis (Module 3); POS profiles with Kruskal Wallis tests and POS trigram patterns (Module 4); dependency based complexity features and a semantic role extractor that produced 23,440 agent, action and patient frames (Module 5); sentence level sentiment and a seven emotion lexicon (Module 6); a comparison of four classifiers on two feature sets with grid search and feature selection (Module 7); and entity analysis with a co-occurrence network and entity sentiment (Module 8). I also built the integrated NewsBotIntelligenceSystem class, tested it on new articles, and wrote the README and documentation.

5. Future Enhancements

The highest value improvements are: fine tuning a transformer such as DistilBERT for classification; training on newer, multi source news and adding a Health category; allowing more than one label for stories that cross topics, which caused most of my 13 classification errors; replacing the small spaCy model with a larger one plus custom rules, since it tagged the Labour party as a person and film titles as money; and wrapping the system in a Streamlit dashboard or API that processes a live news feed and tracks entities over time.

6. Professional Development Impact

This project changed how I think about NLP. Each technique is manageable on its own; the real work is integration, keeping data consistent between steps, and checking whether a result is real or an artifact. Catching the sentiment saturation and the emotion lexicon labeling sports injuries as sadness taught me to question outputs rather than report them. Completing the full pipeline alone also strengthened my project planning. The finished notebook gives me a portfolio piece that

shows an employer I can take raw text to a working, documented system and explain both its value and its limits.